Mini Project | Programming with Python (202044504)

SarthiGo - Car Rental Web App

Search, book and manage car rentals with Django, Tailwind CSS and SQLite

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What is SarthiGo?

A web application for renting cars in Anand. Users search for vehicles that are free on their dates, book one in a few clicks and see their booking history. Staff get their own dashboard to monitor and cancel bookings.

1. **Search**
   Pick dates and seats, see only free cars
2. **Book**
   Driver details, terms, instant confirmation
3. **Manage**
   My Bookings for users, Admin Dashboard for staff

Technology Stack

PY

Backend

Python and Django handle URLs, views, forms, login and booking logic.

UI

Frontend

HTML templates styled with Tailwind CSS (CDN), plus JavaScript date checks.

DB

Database

SQLite, Django's default. No setup needed, stored in one db.sqlite3 file.

Project Structure

```
sarthigo/
  manage.py
  sarthigo/    (settings, urls)
  rentals/
    models.py    Car, Booking
    views.py     page logic
    forms.py     search, signup
    urls.py      routes
    admin.py
    templates/rentals/
      base, home, login,
      register, car_list,
      confirm_booking, ...
```

1 **Model**
Car and Booking tables in SQLite

2 **View**
Python functions: search, book, cancel

3 **Template**
HTML pages that extend base.html

Database Models

Car

- name
- capacity (seats)
- price_per_day
- image_url

Booking

- user (login account)
- car (Car)
- start_date, end_date
- total_cost
- is_approved
- driver name, DOB, license

One car has many bookings. A booking is cancelled by setting is_approved to False, so history is kept.

How a Booking Works



Search

Dates and seats on the home page

Results

Only cars free on those dates

Login

Required before booking

Confirm

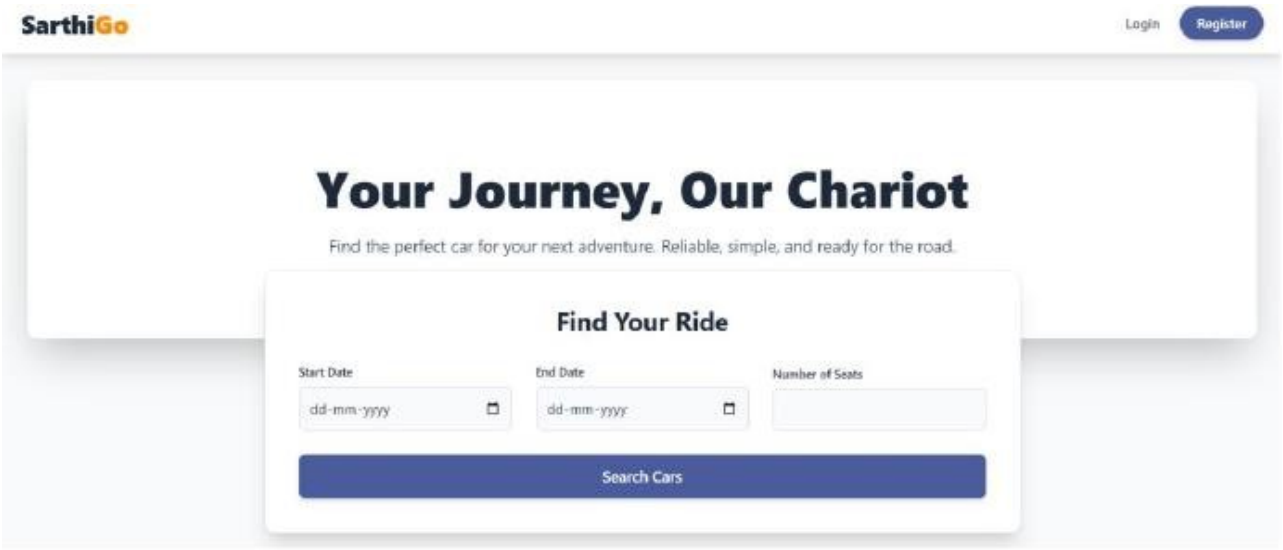
Driver details and terms

Done

Confirmation and My Bookings

Total cost = price per day x number of days (start and end day both count)

Home and Search

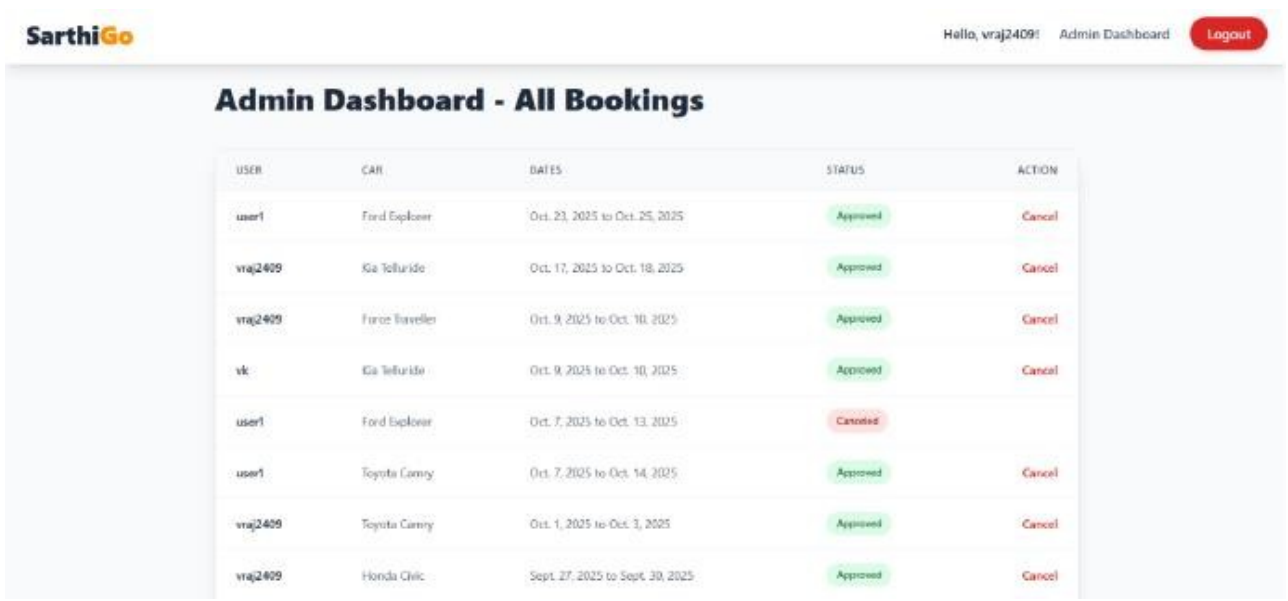


Home page output

- Hero tagline and search form
- Start date, end date, seats
- JavaScript blocks bad dates with an alert
- Server checks the dates again

Admin Dashboard

- Staff-only page
- Table of all bookings from all users
- Approved or Canceled status
- Cancel button with a confirm pop-up



Admin dashboard output

How to Run the Project

1. `pip install -r requirements.txt`
2. `python manage.py makemigrations rentals`
3. `python manage.py migrate`
4. `python manage.py seed_cars`
5. `python manage.py createsuperuser`
6. `python manage.py runserver`
7. Open http://127.0.0.1:8000/

Showing the site

- Run the commands in the folder with manage.py
- Keep the terminal open while presenting
  - Admin: /admin-dashboard/
- Stop server: Ctrl + C

Thank you

SarthiGo - Your Journey, Our Chariot